SMART-LINK: Resource-Constrained Network Resilience

Backup Topology Design, Conflict-Aware Spectrum Assignment, and Secure Relaying with a Reproducible Simulation Suite and Hardware Testbench

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October 2026 | Next.js Graph Playground + 8-Script Python Verification Suite + RP2040 Testbench

Abstract Off-grid emergency and disaster-response networks face a fundamental tension between physical topology resilience and finite radio spectrum. Kruskal's minimum spanning tree (MST) connects dispersed stations at minimum deployment cost, but every tree edge is a single-point bridge whose removal partitions the network. Adding backup links closes protective fundamental cycles, yet simultaneously raises station degree and spatial interference. **SMART-LINK** unifies four engineering layers into a transparent, reproducible framework. *(i) Budgeted submodular cycle coverage:* we protect weighted tree paths under a cardinality budget k, prove that the coverage objective $F(S)$ is normalized, monotone and submodular, and so inherit the classical $(1 - 1/e) \approx 63.2\%$ greedy guarantee against single-link failures; empirically, greedy matches the exhaustive optimum in **96.78%** (871/900) of multi-seed benchmark scenarios. *(ii) Conflict-aware spectrum assignment:* we colour a station-incidence and spatial-midpoint conflict graph $H_c = (E, I)$ over per-link carrier lists $L_e = \mathcal{F} \setminus J_e$ with minimum centre spacing $|f_e - f_{e'}| \geq s$. *(iii) Heterodyne relay translation and RF link-budget physics:* we model analog product-to-sum carrier shifting $f_{\text{out}} = |f_{\text{in}} - f_{\text{LO}}|$ through a 6th-order Butterworth band-pass filter (73–77 kHz), quantifying image suppression (74.3 dB), filter group delay (159.1 µs per hop), the BPSK passband ceiling (≈8 kbit/s) and multi-hop noise accumulation. *(iv) End-to-end payload encryption and relay hop auditing:* endpoint AES-256-GCM is paired with a length-delimited per-relay HKDF + HMAC-SHA256 hop-audit chain τ_i, verified across 1,600 adversarial tamper trials and implemented on a Raspberry Pi Pico (RP2040) control-plane testbench.

Keywords disaster networks; minimum spanning tree; submodular optimization; edge colouring; frequency assignment; heterodyne relay; HMAC hash chain; RP2040.

1 Introduction

When fibre-optic infrastructure is severed by an earthquake, flood or landslide, dispersed emergency-response stations must establish multi-hop radio connectivity with whatever spectrum and hardware survive. Two goals then pull against each other. A cost-minimal backbone has no redundancy, so a single cut isolates part of the network; but every redundant link adds a radio that must be given a carrier that does not collide with its neighbours.

SMART-LINK treats both problems in one pipeline: build a cheap backbone, protect it with a small budget of backup links chosen by a provably near-optimal greedy rule, assign carriers so that conflicting links are separated in frequency, translate carriers at relays with an analog heterodyne stage, and audit the route with a cryptographic hop chain. This paper is the formal companion to the SMART-LINK design paper [8]: it derives the mathematics, reproduces every quoted number with an independent simulation suite, and documents a hardware-in-the-loop testbench. Table 1 reconciles what each artefact implements.

Contributions.

1. A submodularity proof for weighted fundamental-cycle coverage and the resulting $(1 - 1/e)$ greedy bound (Section 4), with an exhaustive-search comparison over 900 scenarios.
2. A list-constrained T-colouring formulation of carrier assignment on a supergraph of the line graph, motivated by the Hadwiger–Nelson problem and hexagonal reuse (Section 5).
3. A quantitative RF reality check of the relay: mixer spectra, filter rejection, group delay, passband data-rate ceiling, amplify-and-forward noise, antenna scaling and self-interference (Sections 6–7).
4. Monte-Carlo resilience tests under link and station failures with on-demand corridor repair (Section 8), and a verified zero-trust hop-audit protocol (Section 9).
5. Reproducible benchmarks to $n = 10{,}000$ stations (Section 11) and an RP2040 testbench (Section 12).

Table 1. Reconciliation of the implemented graph playground, Python simulation suite and hardware scope.

Layer	Mathematical formulation	Next.js graph playground [8]	Python & Pico verification suite (this paper)
1. Backbone & candidates	Symmetrized q-NN within R_c; Kruskal $\min_T \sum_{e\in E_T} c_e$	Implemented in `src/lib/simulation.ts` (5–10,000 stations, Mulberry32 PRNG).	Bit-exact re-implementation in `common.py`; reproduces the Seed-42 ten-station baseline identically.
2. Backup cycle coverage	$\max_{\lvert S\rvert\le k} F(S) = \sum_{e\in \mathrm{Cov}(S)} w_e$, $\mathrm{Cov}(S) = \bigcup_{\ell\in S} P_\ell$	Exact marginal-gain greedy selector + Tarjan bridge enumeration.	Verified against an exact optimal solver over 900 runs (96.78% exact match, worst ratio 0.875) and a 3,000-trial Monte-Carlo.
3. Spectrum assignment	$f_e \in L_e, \lvert f_e - f_{e'}\rvert \ge s$ for all $\{e, e'\} \in I$	Degree-ordered first-fit heuristic; reports unassigned links.	Benchmarked up to $n = 3{,}000$ in spatial mode ($R_I = 0.02, 0.04$) with 1–4 jammed carriers.
4. Outage route recovery	Dijkstra $\min_Q \sum_{e\in Q} c_e$; bypass $E_{\mathrm{rec}} = Q_0 \setminus (E_T \cup S)$	Automatic candidate-corridor activation + prioritized recolouring.	Verified on a 4-station fixture and random multi-station knockouts (1–4 simultaneous losses).
5. Heterodyne relay & RF	$y_{\mathrm{mix}}(t) = s_{\mathrm{in}}(t) \cdot 2\cos(2\pi f_{\mathrm{LO}} t)$; Eqs. (9)–(11)	Specified as engineering design; RF parameters not simulated in the graph UI.	65,536-sample FFT (`04_heterodyne_mixer.py`) + four-panel RF link budget (`05_link_budget.py`).
6. Security & audit chain	AES-256-GCM, Eq. (14); $\tau_i = \mathrm{HMAC}_{K_i}(\mathrm{encode}(\tau_{i-1}, R_i))$	Specified as protocol design with replay and checkpoint rules.	RFC 5869 test vector 1 + 1,600-trial attack matrix (`06_hmac_chain.py`) and RP2040 `pico_node.py`.

2 System Model and Notation

Stations are placed in the unit square by a deterministic 32-bit Mulberry32 generator, so that every experiment is exactly reproducible from its seed. Table 2 collects the notation used throughout.

Table 2. Symbols and nomenclature.

Symbol	Definition and interpretation
V, n	Station set $V = \{0, \ldots, n-1\}$, $n = \lvert V\rvert$; station 0 denotes the origin and station $n-1$ (or a designated node) headquarters (HQ).
p_i, $d(i,j)$	Normalized coordinates $p_i = (x_i, y_i) \in [0,1]^2$, drawn as $x_i = 0.07 + 0.86u_i$, $y_i = 0.08 + 0.84v_i$; Euclidean distance, Eq. (1).
R_c, q, E_0	Candidate radius (default $R_c = 0.55$), neighbour cap (default $q = 12$) and symmetrized q-NN candidate set E_0, $m_0 = \lvert E_0\rvert \le qn$.
c_e, w_e	Link deployment cost $c_e = d(i,j) > 0$ and tree-edge protection weight $w_e \ge 0$ (default $w_e = 1$).
$T = (V, E_T)$, L, S	MST edges ($\lvert E_T\rvert = n-1$), candidate backup pool $L = E_0 \setminus E_T$, and selected backups $S \subseteq L$, $\lvert S\rvert \le k$.
$G = (V, E_T \cup S)$, $\Delta(G)$	Augmented topology with $m = \lvert E_T \cup S\rvert$ active links and maximum station degree $\Delta(G)$.
P_ℓ, H_P	Unique tree path $P_\ell \subseteq E_T$ between the endpoints of backup $\ell = (u,v) \in L$; total cached path size $H_P = \sum_{\ell\in L} \lvert P_\ell\rvert$.
$\mathrm{Cov}(S)$, $F(S)$	Covered tree edges $\mathrm{Cov}(S) = \bigcup_{\ell\in S} P_\ell$ and weighted coverage $F(S) = \sum_{e\in \mathrm{Cov}(S)} w_e$.
$H_c = (E, I)$, $A_{e,e'}$	Conflict graph on active links with conflict edges $I = \{\{e,e'\} : A_{e,e'} = 1\}$ (station incidence, or midpoint distance $\le R_I = 0.04$).
$\mathcal{F}$, J_e, L_e, s	Carrier raster $\mathcal{F} = \{75, 84, 95, 108, 120, 135\}$ kHz, jammed carriers $J_e \subseteq \mathcal{F}$, allowed list $L_e = \mathcal{F} \setminus J_e$, centre spacing $s = 9$ kHz.
B_e, $g_{e,e'}$	Occupied bandwidth and required empty guard gap in Eq. (8).
Q^*, E_{rec}	Minimum-cost surviving route, Eq. (12), and temporary corridor-recovery links, Eq. (13).
f_{in}, f_{LO}, f_{out}, τ_g	Incoming carrier (95 kHz), local oscillator (20 kHz), translated carrier (75 kHz) and band-pass group delay ($\tau_g \approx 159.1$ µs).
K_{enc}, N, AAD, K_i, R_i, τ_i	Endpoint AES-256-GCM key, 96-bit nonce, additional authenticated data, HKDF per-relay audit key, canonical hop record and chained HMAC tag.

3 Backbone Construction and the Bridge Problem

3.1 Symmetrized candidate graph and Kruskal backbone

For every pair of stations within candidate radio range $R_c = 0.55$ we compute the Euclidean deployment cost

$$c_e = d(i,j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \tag{1}$$

Each station i retains its $q = 12$ closest neighbours within R_c, breaking ties deterministically by station ID. An undirected link belongs to E_0 if either endpoint retains the other, giving a symmetrized, radius-constrained q-nearest-neighbour candidate graph $G_0 = (V, E_0)$ with $m_0 = \lvert E_0\rvert \le qn$. Kruskal's algorithm with a disjoint-set (union–find) structure sorts E_0 by

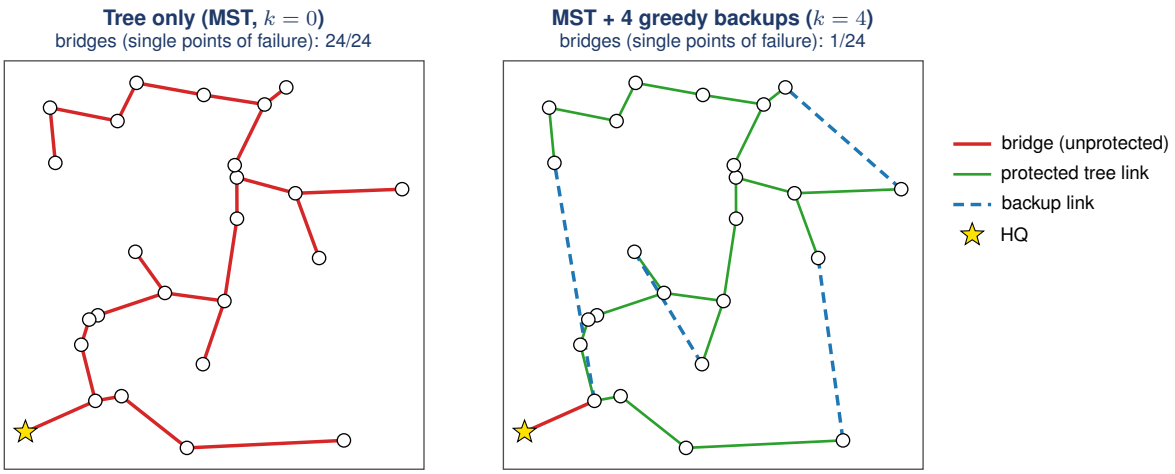


Figure 1. Backup links turn single-point bridges into protected cycles ($n = 25$, Seed 7, regenerated from the Mulberry32 stream). Left: the MST leaves all 24/24 tree edges as vulnerable bridges (red). Right: $k = 4$ greedy backups (dashed) close long fundamental cycles, turning 23 tree edges into protected cycle edges (green) and leaving a single leaf bridge at HQ.

(c_e, u, v) and extracts a minimum-cost spanning tree $T = (V, E_T)$ with exactly $|E_T| = n - 1$ links, or a spanning forest if G_0 is disconnected.

3.2 Every tree link is a bridge

An MST minimizes deployment cost $\sum_{e \in E_T} c_e$ but has zero topological redundancy. An edge whose removal increases the number of connected components is a *bridge* (cut-edge); because a tree has no cycles, *every one* of its $n - 1$ edges is a bridge. A single secondary tremor or equipment fault on, say, link $(0, 2)$ partitions the network into two islands and isolates outer field stations from HQ (Fig. 1, left).

> **Link failures versus station failures**
>
> An individual **link failure** deletes one edge e while all stations remain online. A **station failure** deletes vertex v and *all incident tree and backup links simultaneously*. Protecting every tree edge with a cycle guarantees 2-edge-connectivity (survival of any single link cut) but does **not** eliminate articulation vertices—for example, two cycles sharing a single hub station. Both regimes are tested explicitly in Section 8.

4 Submodular Backup Cycle Coverage

4.1 Fundamental cycles and edge protection

To protect T against single-link cuts with at most k backups $S \subseteq L = E_0 \setminus E_T$, observe that adding a candidate $\ell = (u, v) \in L$ closes a unique fundamental cycle $P_\ell \cup \{\ell\}$, where $P_\ell \subseteq E_T$ is the tree path between u and v. The design problem is

$$\mathrm{Cov}(S) = \bigcup_{\ell \in S} P_\ell, \qquad F(S) = \sum_{e \in \mathrm{Cov}(S)} w_e, \qquad \max_{S \subseteq L,\ |S| \le k} F(S). \tag{2}$$

Lemma 1 (Protection of a tree edge) *For any tree edge $e \in E_T$, the augmented graph $G - e$ remains connected if and only if $e \in \mathrm{Cov}(S)$.*

Proof. Removing e partitions T into two subtrees T_1, T_2. A backup $\ell = (u, v) \in S$ reconnects them if and only if its endpoints lie in opposite subtrees, which holds if and only if the unique tree path P_ℓ crosses the cut edge, i.e. $e \in P_\ell$. Taking the union over $\ell \in S$, $G - e$ stays connected if and only if $e \in \mathrm{Cov}(S)$. ∎

4.2 Diminishing returns and the greedy guarantee

Adding a backup ℓ to a selection $A \subseteq L$ yields the exact marginal gain

$$F(A \cup \{\ell\}) - F(A) = \sum_{e \in P_\ell \setminus \mathrm{Cov}(A)} w_e. \tag{3}$$

Proposition 1 *F is normalized ($F(\emptyset) = 0$), monotone nondecreasing and submodular on 2^L.*

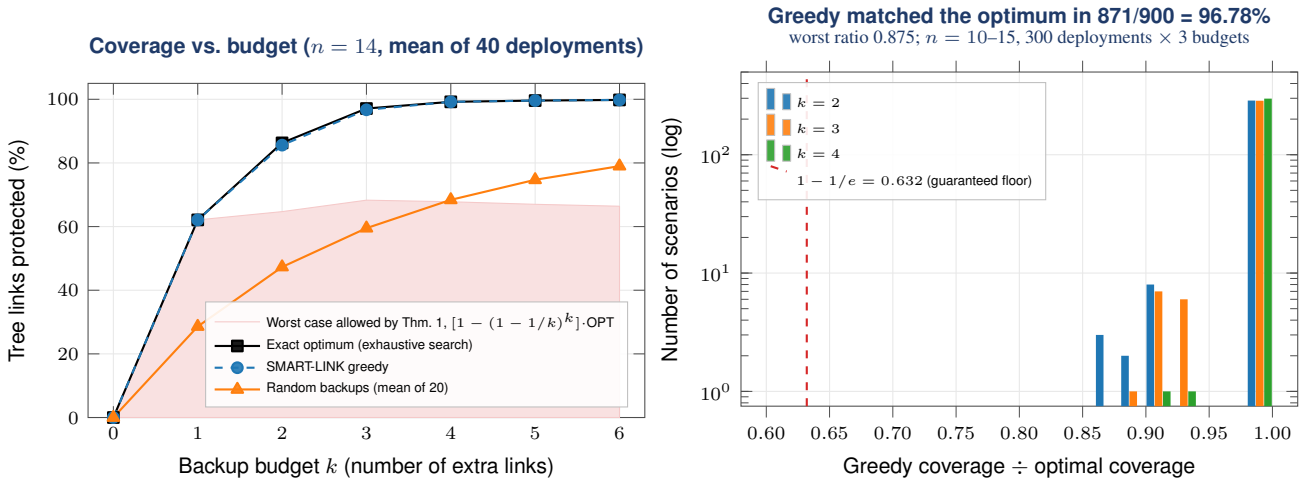


Figure 2. Greedy backup selection versus the exact optimum. Left: mean percentage of tree edges protected versus budget $k \in \{0, \dots, 6\}$ over 40 deployments ($n = 14$), compared with random selection and the theoretical floor. Right: distribution of the greedy/optimal ratio over 900 benchmark scenarios ($n \in [10, 15]$, $k \in \{2, 3, 4\}$).

Proof. Normalization and monotonicity follow from $w_e \geq 0$ and $\mathrm{Cov}(A) \subseteq \mathrm{Cov}(A \cup \{\ell\})$. For $A \subseteq B \subseteq L$ we have $\mathrm{Cov}(A) \subseteq \mathrm{Cov}(B)$, so the uncovered part of the path shrinks, $P_\ell \setminus \mathrm{Cov}(B) \subseteq P_\ell \setminus \mathrm{Cov}(A)$, and by Eq. (3) with $w_e \geq 0$, $F(A \cup \{\ell\}) - F(A) \geq F(B \cup \{\ell\}) - F(B)$. ∎

Theorem 1 (Nemhauser–Wolsey–Fisher [1]) *Let S_g be the set returned by k steps of exact marginal-gain greedy selection and S^* an optimal solution of* (2)*. Then*

$$F(S_g) \;\geq\; \Big[1 - \big(1 - \tfrac{1}{k}\big)^k\Big] F(S^*) \;\geq\; \big(1 - \tfrac{1}{e}\big) F(S^*) \;\approx\; 0.632\, F(S^*). \tag{4}$$

Proof. Let S_t be the greedy set after t steps. By monotonicity and submodularity,

$$F(S^*) - F(S_t) \;\leq\; \sum_{\ell \in S^* \setminus S_t} \big[F(S_t \cup \{\ell\}) - F(S_t)\big]. \tag{5}$$

Since $|S^* \setminus S_t| \leq k$, the best next greedy choice captures at least a $1/k$ fraction of the remaining gap, $F(S^*) - F(S_{t+1}) \leq (1 - 1/k)\,[F(S^*) - F(S_t)]$. Iterating k times from $F(S_0) = 0$ gives (4). ∎

By Lemma 1, $F(S)$ with unit weights counts exactly the tree links that survive any single cut, so Theorem 1 is a resilience guarantee. Figure 2 compares greedy with the exhaustive optimum. On average greedy is indistinguishable from optimal and far above both random selection and the worst case the theorem allows; across 900 scenarios it attains the optimum in 871 (96.78%), with worst ratio 0.875—well above the 0.632 floor.

5 From the Hadwiger–Nelson Problem to Spectrum Assignment

5.1 Plane colouring and hexagonal reuse

The investigation began with the classical Hadwiger–Nelson problem: what is the least number of colours $\chi(\mathbb{R}^2)$ needed to colour the plane so that no two points at unit distance share a colour? It is known that $5 \leq \chi(\mathbb{R}^2) \leq 7$, the lower bound being due to de Grey (2018). Isbell's upper-bound construction tiles the plane with regular hexagons of diameter slightly below 1 in a periodic 7-colour pattern—and that tiling is precisely the geometry of cellular frequency reuse. Valid hexagonal cluster sizes are $N = i^2 + ij + j^2 \in \{1, 3, 4, 7, 9, 12, 13, \dots\}$; the choice $i = 2$, $j = 1$ gives the canonical $N = 7$ cluster with co-channel reuse distance $D = R\sqrt{3N} = \sqrt{21}\,R \approx 4.58R$. With path-loss exponent $\gamma = 4$ and six first-tier interferers the worst-case signal-to-interference ratio is $\mathrm{SIR} \approx (\sqrt{3N})^4/6 = 441/6 = 73.5$ (18.66 dB), above the classic 18 dB threshold (Fig. 3).

5.2 Why point-to-point relays collide

Disaster networks depart from the ideal lattice in two ways: stations are placed irregularly by terrain, and connections are directional point-to-point *links* rather than omnidirectional cells. If two links meeting at a relay transmit and receive on the same carrier simultaneously, the station's own +30 dBm (1 W) transmission is 140 dB—10^{14} times—stronger than a faint incoming signal near the −110 dBm thermal floor, and drowns it. Without multi-channel planning, narrowband radios fall back to half-duplex TDMA: station 1 sends in slot 1 (0–20 ms) while station 2 buffers, and station 2 forwards in slot 2 (20–40 ms). Over long backbones, store-and-forward turn-taking compounds queuing latency at every hop. Separating conflicting links *in frequency* removes the contention.

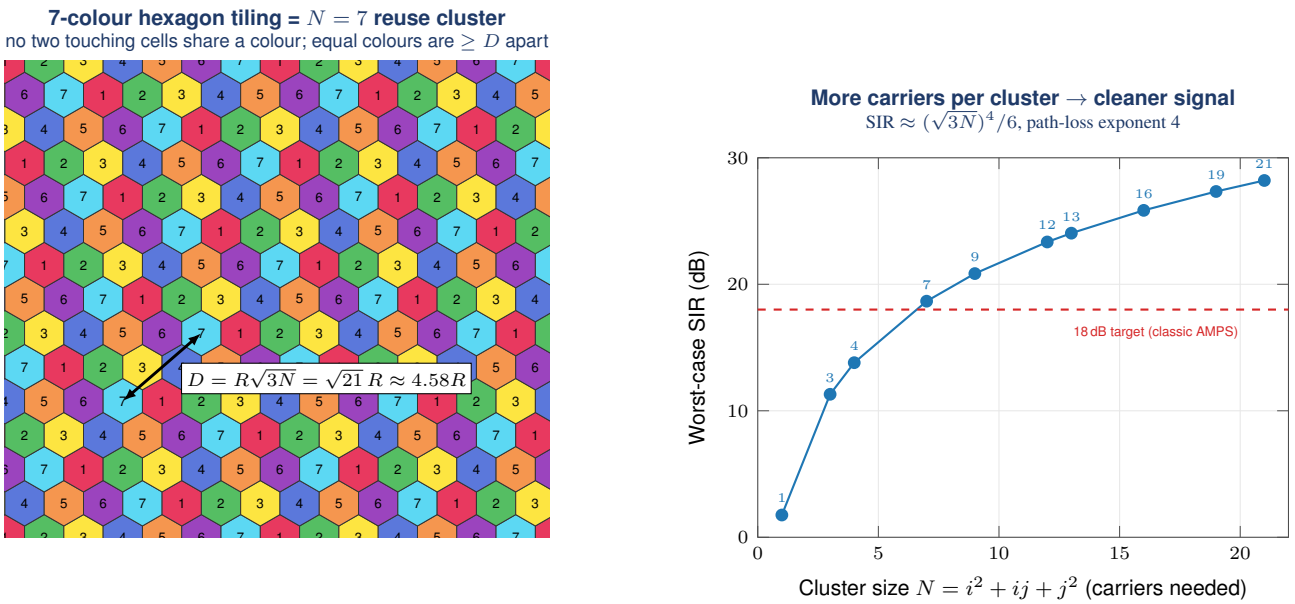


Figure 3. From Hadwiger–Nelson to frequency reuse. Left: Isbell's 7-colour hexagonal tiling (the upper bound $\chi(\mathbb{R}^2) \leq 7$) coincides with the $N = 7$ reuse cluster, here coloured by channel $(q+3r) \bmod 7$ in axial coordinates; co-channel cells are $D = R\sqrt{3N} \approx 4.58R$ apart. Right: first-tier SIR across valid cluster sizes under $\gamma = 4$; point labels give N.

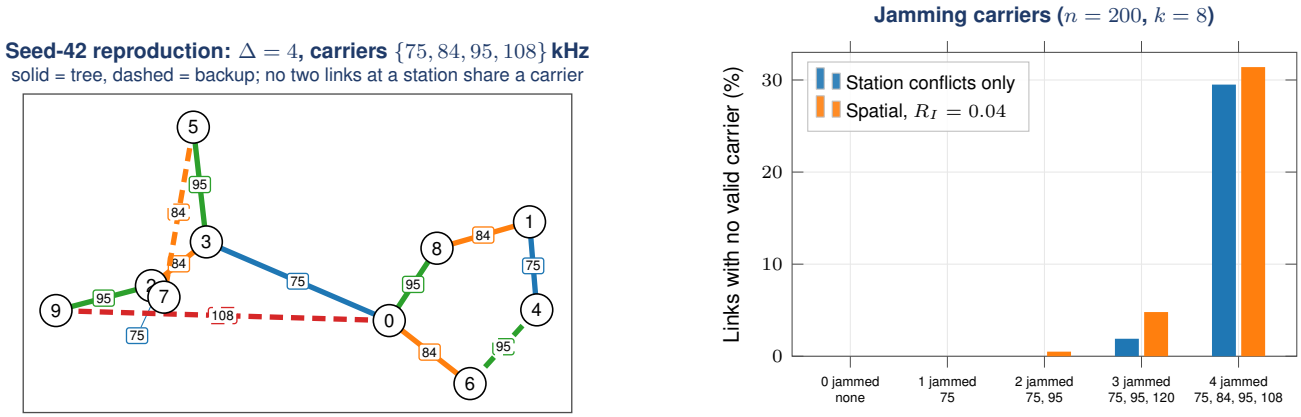


Figure 4. Conflict-aware carrier assignment. Left: bit-exact reproduction of the Seed-42 ten-station baseline ($k = 3$): nine tree links (solid) and three backups (dashed) coloured with four carriers, the provable minimum since $\Delta(G) = 4$. Right: a 200-station network ($k = 8$) as an adversary jams 0–4 of the six raster carriers, under station-only and spatial conflict modes.

5.3 The link conflict graph

Because carriers are assigned to links $e \in E = E_T \cup S$ rather than to towers, SMART-LINK maps $G = (V, E)$ to a conflict graph $H_c = (E, I)$ whose vertices are links. With $m_e = (p_u + p_v)/2$ the midpoint of $e = (u, v)$, two distinct links conflict if

$$A_{e,e'} = \begin{cases} 1 & \text{if } e \cap e' \neq \emptyset \text{ (station incidence), or spatial mode is on and } \|m_e - m_{e'}\|_2 \leq R_I, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

With station-incidence conflicts alone, H_c is the line graph $L(G)$ and Vizing's theorem [3] bounds the edge-chromatic number by $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$. Spatial conflicts (default $R_I = 0.04$), per-link jammed-carrier exclusions and a minimum centre spacing turn the problem into list-constrained T-colouring on a supergraph of $L(G)$:

$$f_e \in L_e = \mathcal{F} \setminus J_e, \qquad |f_e - f_{e'}| \geq s \quad \text{for every } \{e, e'\} \in I, \tag{7}$$

$$|f_e - f_{e'}| \geq \tfrac{B_e + B_{e'}}{2} + g_{e,e'}. \tag{8}$$

SMART-LINK orders links by nonincreasing conflict degree in H_c (ties broken by endpoint IDs) and gives each the lowest carrier in L_e that is at least s from all already-assigned conflicting neighbours, flagging any link that cannot be assigned. Figure 4 shows the bit-exact Seed-42 result and a jamming stress test.

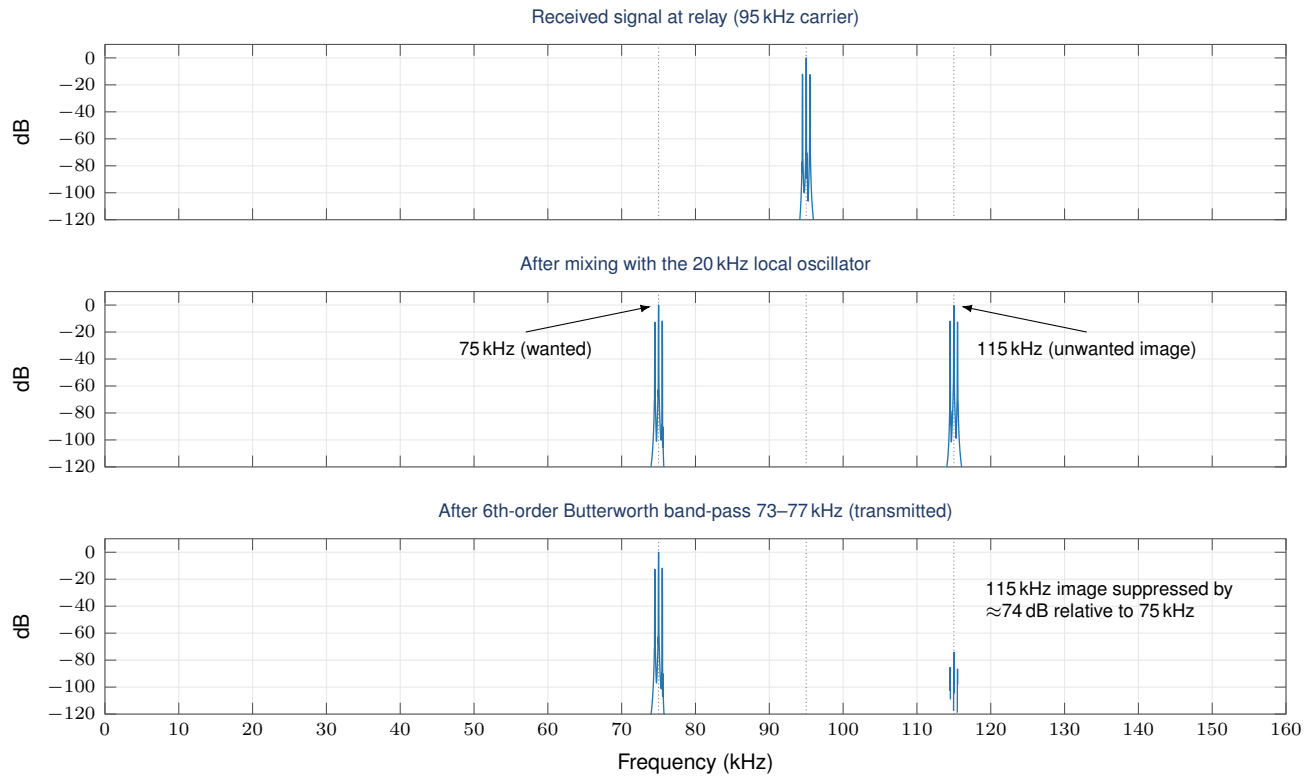


Figure 5. FFT verification of Eq. (9) (65,536 samples, $f_s = 1$ MS/s, Blackman window; recomputed natively). Top: incoming 95 kHz carrier amplitude-modulated by a 500 Hz tone. Middle: mixer output with equal-amplitude products at 75 and 115 kHz. Bottom: after the 73–77 kHz Butterworth band-pass, the 75 kHz signal passes and the 115 kHz image is suppressed by 74.3 dB.

Centre spacing s is not an empty guard band

In Eq. (8), $s = 9$ kHz is a *centre-to-centre* spacing, not a 9 kHz gap. If neighbouring links occupy $B_e = B_{e'} = 4$ kHz, then $(B_e + B_{e'})/2 = 4$ kHz is filled by signal sidebands, leaving an unoccupied guard band $g_{e,e'} = 9 - 4 = 5$ kHz for filter roll-off.

6 Inside the Relay: Heterodyne Mixing and Filter Delay

6.1 Product-to-sum frequency translation

Suppose colouring assigns the incoming link (station 1 $\rightarrow$ relay 2) to $f_{\text{in}} = 95$ kHz and the outgoing link (relay 2 $\rightarrow$ station 3) to $f_{\text{out}} = 75$ kHz. An analog heterodyne relay shifts the carrier without decrypting the payload. Let $s_{\text{in}}(t) = a(t)\cos(2\pi f_{\text{in}} t + \phi(t))$ and multiply by a local oscillator $s_{\text{LO}}(t) = 2\cos(2\pi f_{\text{LO}} t)$ with $f_{\text{LO}} = 20$ kHz. The identity $2\cos A\cos B = \cos(A-B) + \cos(A+B)$ gives

$$y_{\text{mix}}(t) = a(t)\cos\big(2\pi(f_{\text{in}} - f_{\text{LO}})t + \phi(t)\big) + a(t)\cos\big(2\pi(f_{\text{in}} + f_{\text{LO}})t + \phi(t)\big), \tag{9}$$

i.e. $Y_{\text{mix}}(f) = S_{\text{in}}(f - f_{\text{LO}}) + S_{\text{in}}(f + f_{\text{LO}})$, where the factor 2 in s_{LO} preserves unit passband amplitude. Mixing therefore produces the wanted difference carrier at 75 kHz and an unwanted sum image at 115 kHz, which a band-pass filter must remove (Fig. 5).

6.2 Adjacent-channel rejection and group delay

Filter rejection and group delay must be measured rather than assumed. Evaluating the 6th-order (order-3 band-pass prototype) Butterworth filter with `scipy.signal.sosfreqz` and its phase derivative establishes:

- **Adjacent-carrier rejection:** 37.9 dB at the nearest raster neighbour (84 kHz), 57.4 dB at 95 kHz, 69.3 dB at 108 kHz, and 73.8–74.3 dB at the 115 kHz sum image.
- **Non-zero group delay:** any causal, sharp band-pass filter stores energy in its reactive stages. At 75 kHz the group delay is $\tau_g = 159.1$ µs per relay hop, peaking above 200 µs near the 73 and 77 kHz band edges. Analog translation is thus $\approx$125$\times$ faster than a 20 ms TDMA slot, but not instantaneous (Fig. 6).

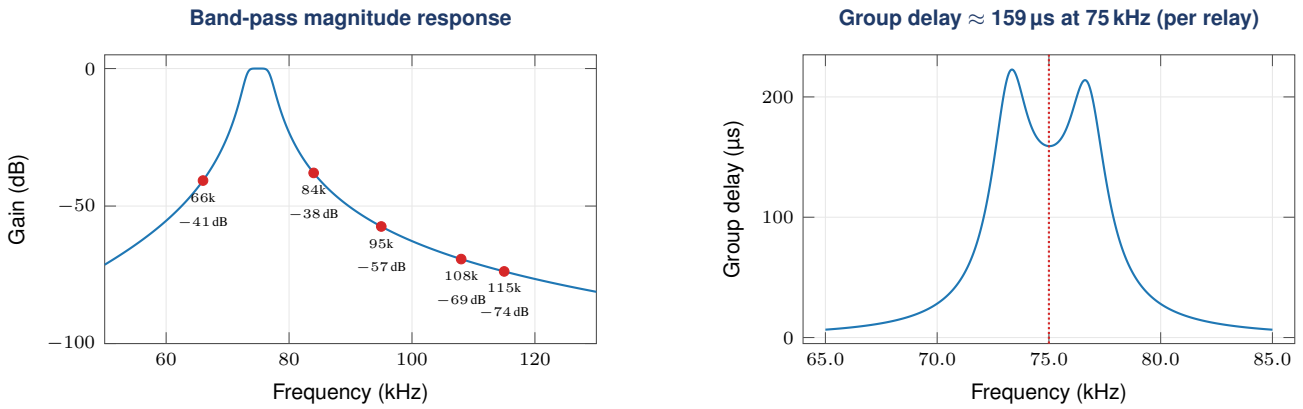


Figure 6. The relay filter is not instantaneous. Left: magnitude response of the 6th-order 73–77 kHz Butterworth band-pass with attenuation at neighbouring raster frequencies. Right: passband group delay (159.1 µs at 75 kHz), the physical filter latency added at each hop.

7 Physical-Layer RF Analysis

Full-duplex and multi-link operation depend on residual transmitter leakage and on the signal-to-interference-plus-noise ratio,

$$P_{\text{residual}} = P_{\text{tx}} - A_{\text{antenna}} - A_{\text{filter}} - A_{\text{cancellation}} \quad \text{(dBm, dB)}, \tag{10}$$

$$\text{SINR}_e = \frac{P_e\, g_e}{N_0 B_e + \sum_{j \neq e} P_j\, g_{j \to e}\, \eta_{j,e}}, \qquad C_e = B_e \log_2(1 + \text{SINR}_e). \tag{11}$$

To keep narrowband low-IF benchmarks distinct from wideband tactical-radio specifications, we ran four quantitative evaluations (Figs. 7 and 8).

Passband data-rate ceiling. By the Shannon limit in Eq. (11), a 4 kHz filter cannot carry wideband 128 kbit/s or 16 Mbit/s waveforms: 128 kbit/s in 9 kHz already needs $\text{SINR} \geq 42.8$ dB, and 16 Mbit/s needs megahertz of UHF/SHF spectrum. Sweeping BPSK symbol rates through the 73–77 kHz filter gives 0% bit errors up to 8 kbit/s, then severe inter-symbol interference (20.4% BER at 12 kbit/s) once symbol bandwidth exceeds the passband.

Amplify-and-forward noise. Unlike decode-and-forward relays, which regenerate clean bits, analog heterodyne relays amplify incoming thermal noise at every stage. Over h identical hops of per-hop SNR_1, $\text{SNR}_{\text{total,dB}} = \text{SNR}_{1,\text{dB}} - 10\log_{10} h$: 10 dB lost over 10 hops, 20 dB over 100.

Quarter-wave antenna scaling. At 75 kHz (LF, $\lambda = c/f = 4{,}000$ m) a quarter-wave whip is $\lambda/4 = 1{,}000$ m tall. The 75–135 kHz raster is therefore a mathematical raster or internal IF/ultrasonic test stage; field radios up-convert to 433 MHz LoRa/ISM ($\lambda/4 = 17.3$ cm) or 2.4 GHz ($\lambda/4 = 3.1$ cm).

Same-band full-duplex leakage. With $P_{\text{tx}} = +30$ dBm and 85 dB aggregate suppression, Eq. (10) leaves $P_{\text{residual}} = -55$ dBm—still 55 dB above a -110 dBm noise floor. Reaching the floor needs 140 dB total isolation [2, 4], or frequency-separated FDD across distinct carriers.

8 Outages, Articulation Stations and Corridor Repair

8.1 Distance-optimal routing and on-demand corridor recovery

In normal operation, or after minor cuts, SMART-LINK computes a minimum-distance (or minimum-hop BFS) route between origin a and HQ b over the surviving carrier-assigned links E_{usable} with binary-heap Dijkstra in $O((n+m)\log n)$:

$$Q^* \in \operatorname*{arg\,min}_{Q:\, a \rightsquigarrow b,\ Q \subseteq E_{\text{usable}}} \sum_{e \in Q} c_e. \tag{12}$$

When a severe multi-link or articulation-station outage destroys every pre-deployed path in $E_T \cup S$, the engine attempts on-demand repair: (1) failed stations and links are removed from the full candidate pool E_0; (2) Dijkstra finds a shortest surviving candidate corridor Q_0; (3) the unchosen candidate links along it are temporarily activated,

$$E_{\text{rec}} = Q_0 \setminus (E_T \cup S); \tag{13}$$

and (4) conflict-aware carrier assignment is re-run with E_{rec} first, so the bypass reserves clean spectrum. Activating E_{rec} uses spare candidate resources outside the static budget k, and is undone once failed nodes recover.

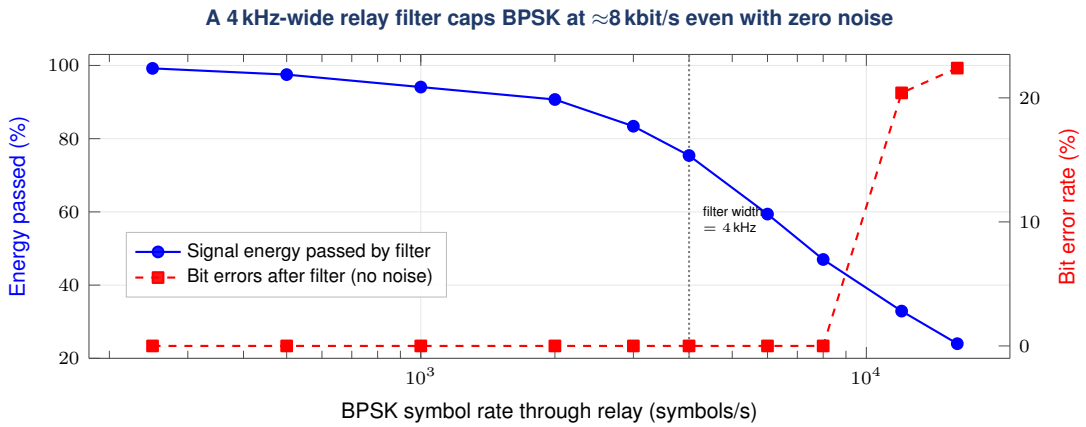


Figure 7. BPSK symbol-rate sweep through the 6th-order 73–77 kHz (4 kHz) Butterworth band-pass with delay-correlation synchronization: error-free up to 8 kbit/s, sideband clipping above 10 kbit/s. The 128 kbit/s and 16 Mbit/s channels sometimes quoted for such links cannot fit.

8.2 Monte-Carlo disaster stress tests

To look beyond single cuts, `02_failure_montecarlo.py` subjects a 40-station network (Seed 3) to 3,000 independent trials per point under (a) independent link failure with probability $p \in [0, 30\%]$ and (b) destruction of 1–4 whole stations. At $p = 5\%$, whole-network survival rises from 13.5% (tree only) to 74.3% (greedy, Fig. 9). With $k = 8$, pure tree-coverage greedy stops at five backups because they already cover every tree edge (zero marginal gain left), whereas random $k = 8$ adds three extra chords that help under extreme $p > 15\%$ multi-link cuts—coverage of single cuts is not the same objective as survival of many. Station losses (Fig. 10) separate the regimes further: in the bare tree 31 of 39 non-HQ stations are articulation points.

9 Zero-Trust Relaying: AES-256-GCM and Chained HMAC

9.1 Endpoint payload privacy versus control-plane hop auditing

Relays in a disaster zone may be physically exposed or untrusted, so SMART-LINK separates endpoint payload encryption from per-hop route auditing. The origin v_0 encrypts plaintext M with AES-256-GCM [5] under a key K_{enc} known only to origin and HQ, a unique 96-bit nonce N (32-bit sender/epoch prefix + 64-bit persistent packet counter) and immutable additional authenticated data (AAD binding source, destination, session ID and sequence number):

$$(C, T_{\text{GCM}}) = \text{GCM-Enc}_{K_{\text{enc}}}(M, N, \text{AAD}), \tag{14}$$

$$\tau_i = \text{HMAC-SHA256}_{K_i}\big(\text{encode}(\tau_{i-1}, R_i)\big). \tag{15}$$

An analog mixer cannot append bytes by itself; instead each relay's digital control processor parses the unencrypted routing header and appends a length-delimited canonical hop record R_i—binding session ID, packet sequence, relay ID v_i, hop index i, outgoing carrier f_{out} and a SHA-256 digest of the immutable ciphertext envelope—signed with a per-relay audit key K_i derived via HKDF [6, 7].

9.2 Empirical verification

In `06_hmac_chain.py` the HKDF-SHA256 implementation is first checked against RFC 5869 Test Case 1 (matching the 42-byte OKM bit for bit); we then run 200 adversarial trials for each of eight attack scenarios on a five-hop route (1,600 verifications). Modifying any carrier, deleting an interior hop, swapping two hops or swapping the ciphertext digest is caught every time (Fig. 11). The experiment also exhibits two boundaries of the design: (1) *tail truncation*—dropping the last hops and presenting the earlier valid tag τ_{i-1}—goes undetected unless HQ checks an expected final hop count or terminal checkpoint, which restores 100% detection; and (2) a *compromised terminal relay* can sign a false carrier inside its own record R_i with its legitimate key K_i. Verification cost is strictly linear in the hop count, ≈9.8 µs per hop including per-hop HKDF derivation (Fig. 12).

10 Algorithmic Complexity

Table 3 lists the data structures and asymptotic cost of each pipeline stage. Two practical caveats matter: the total cached path size H_P can approach $m_0 n$, so the greedy stage dominates for large n; and under dense spatial conflicts the number of conflict pairs can reach $J = O(m^2)$.

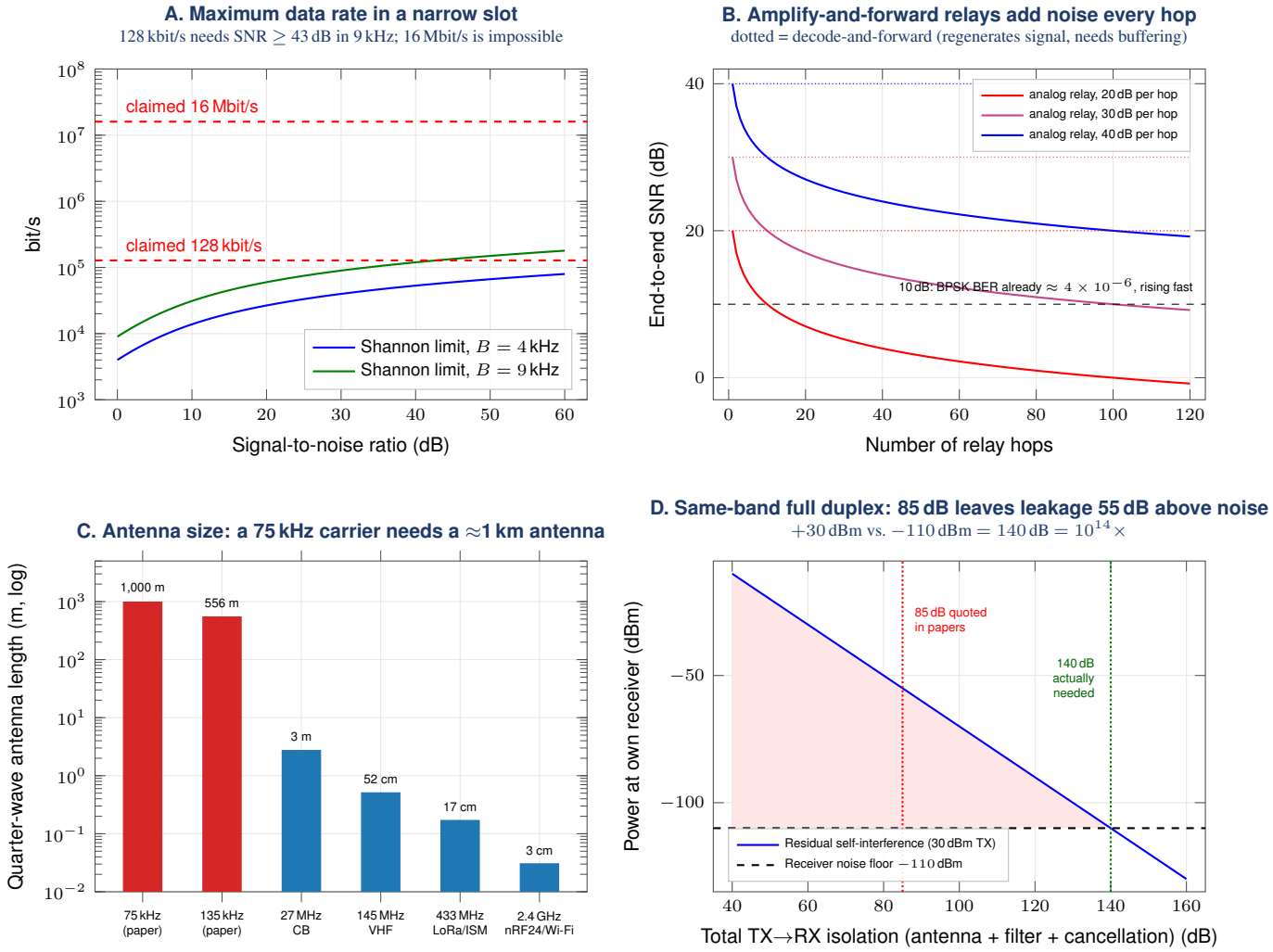


Figure 8. RF reality check: (A) Shannon capacity of 4 and 9 kHz channels; (B) amplify-and-forward SNR loss over h hops versus regenerative decode-and-forward; (C) quarter-wave antenna length across bands; (D) the self-interference budget of Eq. (10) (+30 dBm TX versus a −110 dBm floor).

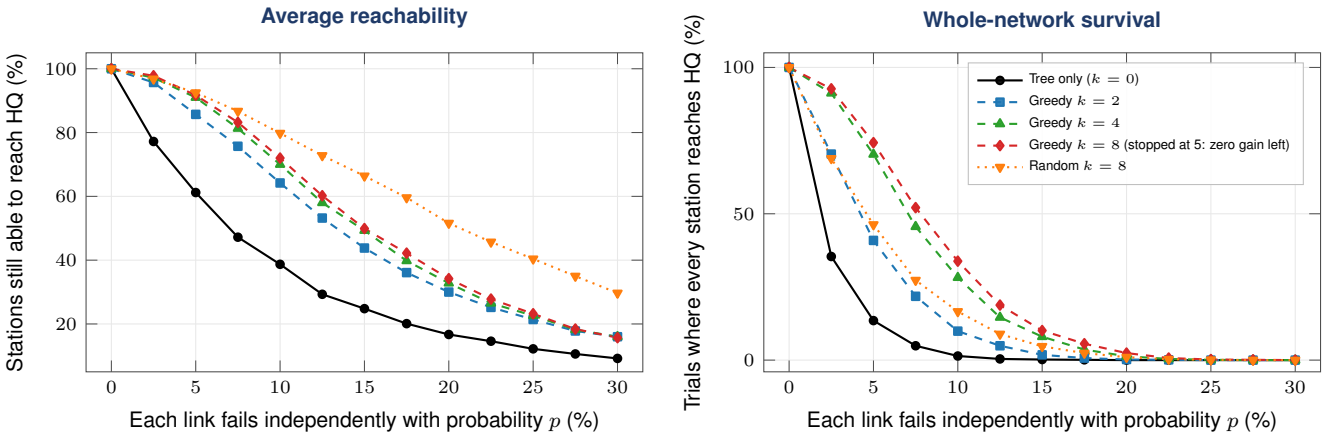


Figure 9. Monte-Carlo link-failure test ($n = 40$, 3,000 trials per point). Left: mean fraction of stations that can still reach HQ. Right: fraction of trials in which *every* station reaches HQ. At $p = 5\%$ survival rises from 13.5% (tree only) to 74.3% (greedy).

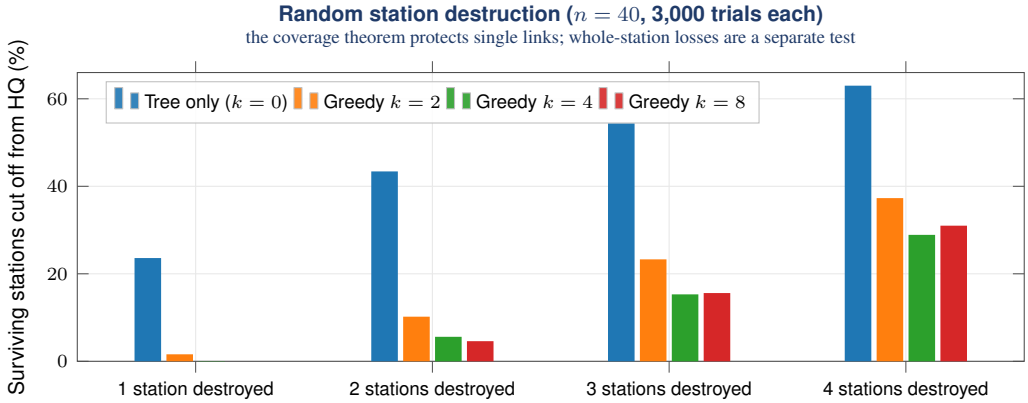


Figure 10. Mean percentage of surviving stations cut off from HQ when 1–4 whole stations are destroyed. In the unaugmented tree, losing one random station cuts off 23.6% of survivors; greedy cycle backups reduce this to 1.60% ($k = 2$), 0.07% ($k = 4$) and 0.00% ($k = 8$). Multi-station losses show why vertex connectivity and on-demand repair (E_{rec}) remain essential.

11 Reproducible Software Benchmarks

All benchmarks use the deterministic Mulberry32 generator with $R_c = 0.55$, $q = 12$, $s = 9\,\text{kHz}$, $R_I = 0.04$ and spatial mode off by default. On the Seed-42 ten-station baseline ($n = 10$, $k = 3$), the TypeScript engine (`src/lib/simulation.ts`) and the independent Python engine (`common.py`) both produce 35 candidate links, 9 tree links, greedy backups $(4, 6)$, $(5, 7)$, $(0, 9)$ with marginal gains $[4, 3, 2]$, 9/9 protected tree edges, no remaining bridges, $\Delta(G) = 4$, and the four carriers $\{75, 84, 95, 108\}$ kHz—provably minimal since $\chi'(G) \geq \Delta(G) = 4$. The figures in this paper were regenerated from the same stream and reproduce these values exactly. Random $k = 3$ selection across 20 seeds (1000–1019) protects a mean of 5.65/9 edges (range 4–8). Table 4 reports scaling to 10,000 stations.

A log–log fit of the $n \geq 1{,}000$ timings gives $t \propto n^{1.54}$, faster than $n \log n$ because fundamental path lengths, and hence H_P, grow with n (Fig. 13, left). Holding $k = 32$ fixed while n grows to 10,000 drops tree-edge protection from 100% to 33.7–36.3% (right): resilience budgets must scale with network size, e.g. $k \approx 0.15n$. Spectrum behaves similarly (Fig. 14): with station-incidence conflicts only, six carriers always suffice, but spatial midpoint conflicts ($R_I = 0.04$) leave 7.1% of links unassigned at $n = 1{,}000$ and 51.3% at $n = 3{,}000$ as link density grows inside $[0, 1]^2$.

12 Raspberry Pi Pico Hardware-in-the-Loop Testbench

We distinguish physical RF full-duplex cancellation, which needs specialized RF front-end hardware, from the Raspberry Pi Pico (RP2040) testbench (`pico/pico_node.py`). Using three or four Pico boards, the testbench executes the real digital control plane and makes frequency translation audible:

- **Embedded HKDF + chained HMAC-SHA256.** Each Pico derives its relay key K_i by RFC 5869 HKDF, parses incoming JSON packets over UART1 (115,200 baud), computes Eq. (15) with a canonical fixed-order byte encoding, and logs processing latency in microseconds (`time.ticks_us()`).

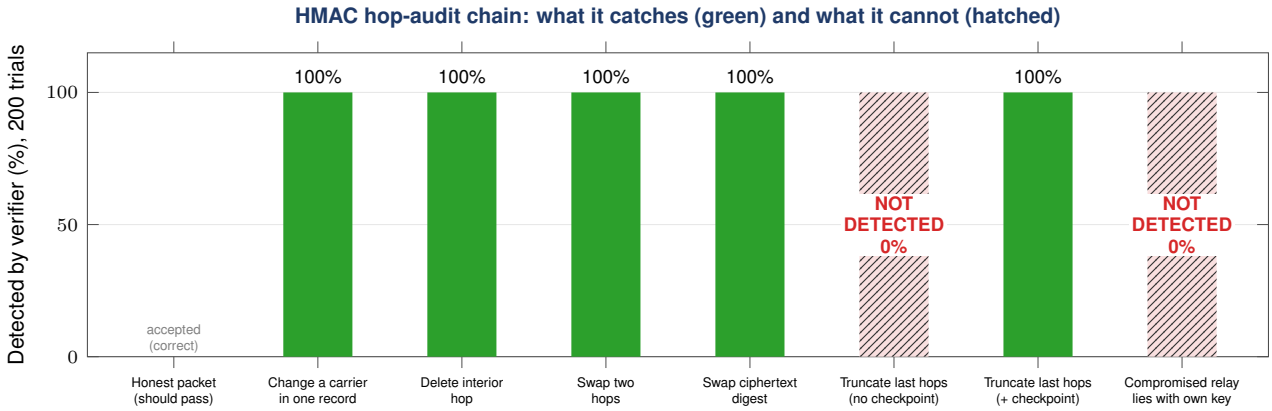


Figure 11. Detection rate across eight scenarios (200 trials each). Tampering with any record, hop order or ciphertext digest is always detected. Tail truncation is detected only with a terminal hop-count checkpoint, and a compromised relay signing a false value with its own key cannot be detected by the chain alone.

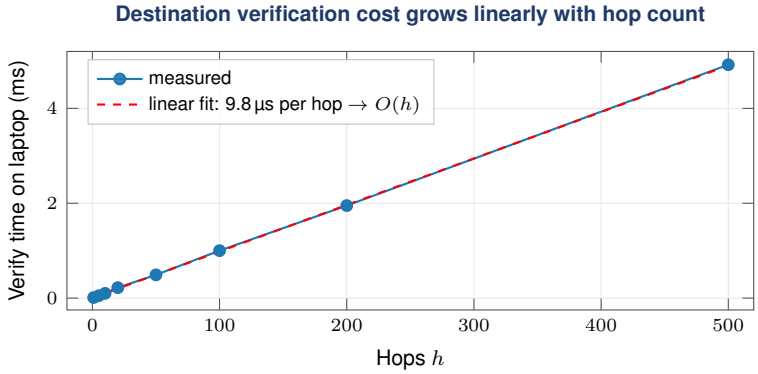


Figure 12. Destination verification time versus hop count $h \in [1, 500]$, confirming $O(h)$ verification at ≈9.8 µs per hop including per-hop HKDF key derivation.

- **Audible carrier translation.** Each Pico drives a piezo buzzer on GP15 at its assigned carrier scaled by $1/100$ into the audio band (75 kHz→750 Hz, 84→840 Hz, 95→950 Hz, 108→1,080 Hz), so each hop's frequency shift can be heard.
- **Interactive tamper button (GP13).** Holding the button on any relay injects a live man-in-the-middle attack (mutating the outgoing carrier metadata to $f_{\text{out}} = 1350$ after HMAC tagging). The destination Pico immediately extinguishes its green verification LED (GP14) and reports `verified=0` over USB serial.

The relay audit loop at the core of `pico_node.py` is:

```
def encode(prev, rec):  # canonical fixed-order encoding (immune to dict key ordering)
    body = '{}|{}|{}|{}|{}|{}'.format(rec['s'], rec['q'], rec['r'], rec['h'], rec['f'], rec['ct']).encode()
    return len(prev).to_bytes(4, 'big') + prev + len(body).to_bytes(4, 'big') + body

# inside the RELAY loop, on receiving a packet over UART1:
t_start = time.ticks_us()
rec = {'s': SESSION, 'q': pkt['seq'], 'r': NODE_ID, 'h': len(pkt['hops']) + 1, 'f': CARRIER_HZ, 'ct': pkt['ct'
    ][:16]}
tau = hmac_sha256(K, encode(ubinascii.unhexlify(pkt['tau']), rec))
if not evil.value(): rec['f'] = 1350   # GP13 tamper switch pressed -> corrupt carrier after signing
rec['us'] = time.ticks_diff(time.ticks_us(), t_start)
pkt['hops'].append(rec); pkt['tau'] = ubinascii.hexlify(tau).decode(); send(pkt)
```

From UART jumper wires to true multi-frequency RF

To demonstrate wireless multi-channel failover, an nRF24L01+ (2.4 GHz SPI transceiver, about INR 150) or SX1278 LoRa (433 MHz) module can be attached to each Pico's SPI0 pins (GP16–GP19), mapping SMART-LINK's assigned carrier f_e directly to `nrf.set_channel(ch)`.

Table 3. Pipeline stages, data structures and complexity.

Stage	Algorithm & data structure	Time	Space	Notes
1. Candidate pruning	Spatial grid / k-d tree + symmetrized q-NN cap ($q = 12$)	$O(n^2)$ worst; fast grid average	$O(n + m_0)$, $m_0 \le qn$	Restricted candidate set, not the full radius graph.
2. Backbone tree	Kruskal MST, path-compressed union–find	$O(m_0 \log m_0)$	$O(n + m_0)$	Spanning forest if G_0 is disconnected.
3. LCA path cache	Rooted parent/depth arrays + endpoint-to-LCA climb	$O(n + m_0 + H_P)$	$O(n + m_0 + H_P)$	$H_P = \sum_{\ell \in L} \lvert P_\ell \rvert \le m_0 n$.
4. Greedy backups	k exact marginal-gain scans over L	$O(k \cdot H_P)$	$O(\lvert S \rvert + n)$	Stops early when the best gain is 0.
5. Bridge detection	Iterative Tarjan low-link DFS	$O(n + m)$	$O(n + m)$	Finds uncovered tree edges.
6. Carrier colouring	Degree-ordered first-fit over $r = \lvert \mathcal{F} \rvert$ carriers and $J = \lvert I \rvert$ conflicts	$O(m \log m + r(m + J))$	$O(n + m + \Delta(H_c))$	Dense spatial conflicts: $J = O(m^2)$.
7. Route & repair	Binary-heap Dijkstra (or BFS) + optional E_{rec} pass	$O((n + m) \log n)$	$O(n + m)$	$O(1)$ table lookup $\neq$ physical failover latency.

Table 4. Multi-seed scaling (Node.js v24.14.0, AMD Ryzen 7 260).

Stations n	Budget k	Median (ms)	Min–max (ms)	Protected links	% protected	Interpretation
10	3	0.86	0.24 – 1.98	9 – 9 / 9	100.0	Sub-millisecond; all 9 bridges eliminated.
100	32	5.08	3.55 – 5.27	99 – 99 / 99	100.0	$k = 32$ covers all 99 tree edges.
1,000	32	83.01	60.37 – 169.98	726 – 751 / 999	72.7 – 75.2	Interactive (< 0.1 s); ≈26% bridges remain.
5,000	32	919.27	705.15 – 1,505.54	2,130 – 2,327 / 4,999	42.6 – 46.5	Fixed $k = 32$ covers under half the tree.
10,000	32	2,951.94	2,364.43 – 4,279.05	3,373 – 3,632 / 9,999	33.7 – 36.3	≈65% remain bridges unless $k \propto n$.

Table 5. Four-Pico daisy-chain wiring and pinout.

Board / role	Node ID & carrier	UART1 data wiring	Peripherals (GP13–15)	Action
Pico 0: origin	`NODE_ID=0`, 750 Hz	GP4 (TX) → Pico 1 GP5 (RX); common GND	GP14 TX LED; GP15 750 Hz buzzer	Generates packet + initial session tag τ_0 every 200 ms.
Pico 1: relay 1	`NODE_ID=1`, 840 Hz	GP5 (RX) ← Pico 0 GP4; GP4 (TX) → Pico 2 GP5	GP13 tamper switch to GND; GP14 hop LED; GP15 840 Hz	Appends R_1, updates τ_1; hold GP13 to tamper.
Pico 2: relay 2	`NODE_ID=2`, 950 Hz	GP5 (RX) ← Pico 1 GP4; GP4 (TX) → Pico 3 GP5	GP13 tamper switch to GND; GP14 hop LED; GP15 950 Hz	Appends R_2, updates τ_2, measures hop time.
Pico 3: HQ	`NODE_ID=99`, USB serial	GP5 (RX) ← Pico 2 GP4; USB to laptop	GP14 green verify LED; CSV `seq,hops,verified,us`	Recomputes the $O(h)$ chain; lights LED iff valid.

13 Limitations

- **Restricted candidate graph.** Coverage guarantees and MST optimality hold relative to the symmetrized q-NN graph G_0 ($R_c = 0.55$, $q = 12$); full radius graphs or terrain-obstructed deployments can yield different topologies.
- **Single-link versus multi-failure resilience.** Theorem 1 protects weighted tree edges against single-link cuts. Simultaneous multi-link cuts, articulation-station losses and correlated disasters require higher vertex connectivity or spare candidate links via on-demand repair (E_{rec}).
- **Centralized recolouring.** The playground recomputes carrier assignments globally; field deployment requires a distributed control channel and a transition handshake.
- **RF and cryptographic deployment.** The suite verifies the FFT and Butterworth response, the link-budget equations and the HKDF + HMAC tamper matrix; operational full-duplex cancellation (> 140 dB from $+30$ dBm TX to a -110 dBm floor), key provisioning, persistent nonce state and terminal hop-count checkpoints remain deployment requirements.

14 Conclusion

SMART-LINK shows that a disaster network's cost, resilience, spectrum and trust can be engineered in one transparent pipeline. Cycle coverage is submodular, so a simple greedy rule protects at least 63.2% of what an optimal budgeted design could—and in practice matches it in 96.78% of scenarios. Carrier assignment is a list T-colouring that meets Vizing's lower bound on small networks but saturates a fixed raster under dense spatial interference; analog relaying is fast (159.1 µs per

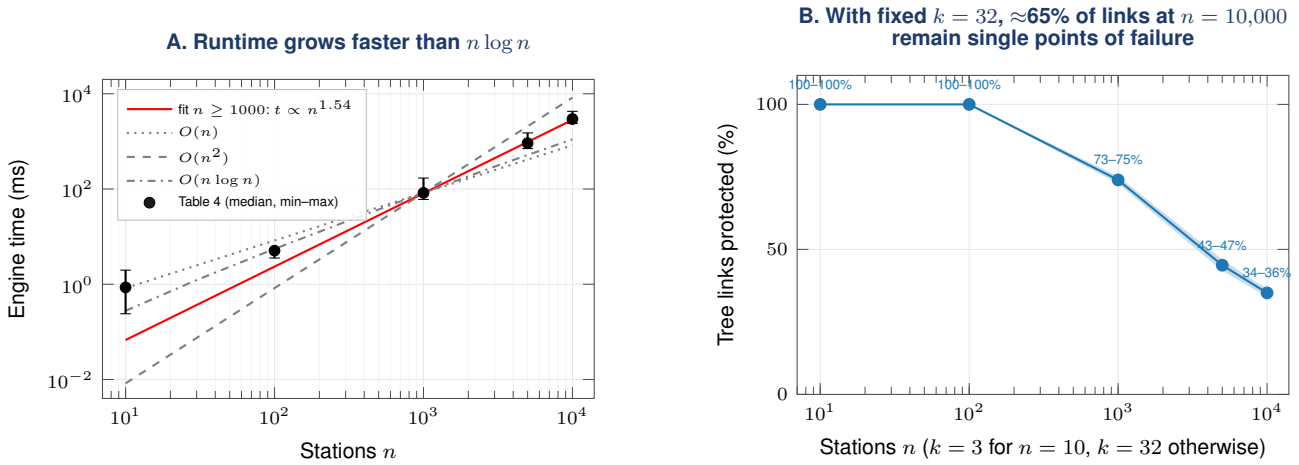


Figure 13. Re-analysis of the scaling table. Left: log–log fit of Table 4 execution times, showing empirical $t \propto n^{1.54}$ for $n \geq 1{,}000$ against $O(n)$, $O(n \log n)$ and $O(n^2)$ references anchored at $n = 1{,}000$. Right: with $k = 32$ fixed, tree-edge protection falls from 100% to 33.7–36.3% (band = min–max over seeds).

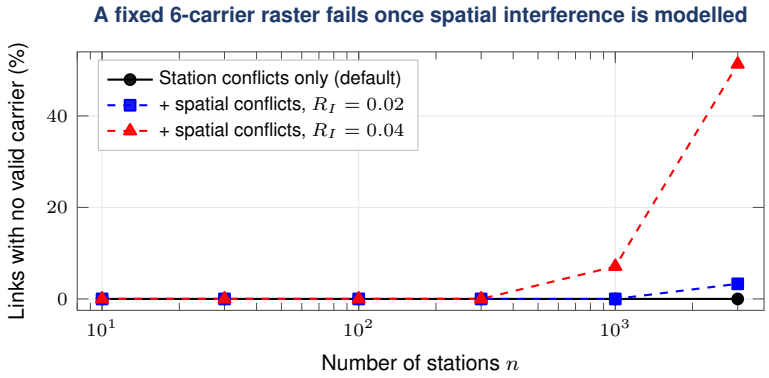


Figure 14. Percentage of links left without a valid carrier as n scales from 10 to 3,000. Six carriers suffice for station-incidence conflicts; spatial midpoint conflicts at $R_I = 0.04$ leave 7.1% unassigned at $n = 1{,}000$ and 51.3% at $n = 3{,}000$.

hop) but capped at ≈8 kbit/s in a 4 kHz passband and accumulates noise; and an HMAC hop chain catches every in-path modification provided HQ checks the terminal hop count. Each of these claims is reproduced by the open simulation suite and the RP2040 testbench, and each comes with an explicit boundary of validity.

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A Equation Summary

Eq.	Layer	Statement and scope
(1)	Candidate graph	$d(i,j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$, retaining symmetrized top-q neighbours within R_c.
(2)	Cycle coverage	$\mathrm{Cov}(S) = \bigcup_{\ell \in S} P_\ell$, $F(S) = \sum_{e \in \mathrm{Cov}(S)} w_e$, $\max_{\lvert S\rvert \le k} F(S)$.
(3)	Marginal gain	$F(A \cup \{\ell\}) - F(A) = \sum_{e \in P_\ell \setminus \mathrm{Cov}(A)} w_e$, nonincreasing in $A \Rightarrow$ submodular.
(4)	Greedy guarantee	$F(S_g) \ge [1 - (1 - 1/k)^k] F(S^*) \ge (1 - 1/e) F(S^*)$ for fixed L and $w_e \ge 0$.
(5)	Submodular gap	$F(S^*) - F(S_t) \le \sum_{\ell \in S^* \setminus S_t} [F(S_t \cup \{\ell\}) - F(S_t)]$.
(6)	Conflict graph	$A_{e,e'} = 1$ if $e \cap e' \neq \emptyset$ or (spatial mode and $\lVert m_e - m_{e'} \rVert_2 \le R_I$).
(7)	Carrier separation	$f_e \in L_e = \mathcal{F} \setminus J_e$, $\lvert f_e - f_{e'} \rvert \ge s$ for every $\{e, e'\} \in I$.
(8)	Guard-band rule	$\lvert f_e - f_{e'} \rvert \ge (B_e + B_{e'})/2 + g_{e,e'}$.
(9)	Heterodyne mixer	$y_{\mathrm{mix}} = a \cos(2\pi(f_{\mathrm{in}} - f_{\mathrm{LO}})t + \phi) + a \cos(2\pi(f_{\mathrm{in}} + f_{\mathrm{LO}})t + \phi)$.
(10)	Leakage budget	$P_{\mathrm{residual}} = P_{\mathrm{tx}} - A_{\mathrm{antenna}} - A_{\mathrm{filter}} - A_{\mathrm{cancellation}}$.
(11)	Physical SINR	$\mathrm{SINR}_e = P_e g_e / (N_0 B_e + \sum_{j \neq e} P_j g_{j \to e} \eta_{j,e})$; $C_e = B_e \log_2(1 + \mathrm{SINR}_e)$.
(12)	Shortest routing	$Q^* \in \arg\min_{Q \subseteq E_{\mathrm{usable}}} \sum_{e \in Q} c_e$ (or $\min \lvert Q \rvert$ in BFS mode).
(13)	Corridor repair	$E_{\mathrm{rec}} = Q_0 \setminus (E_T \cup S)$, prioritized during recolouring after severe outages.
(14)	Endpoint GCM	$(C, T_{\mathrm{GCM}}) = \text{GCM-Enc}_{K_{\mathrm{enc}}}(M, N, \mathrm{AAD})$, unique 96-bit nonce, 128-bit tag.
(15)	Relay HMAC chain	$\tau_i = \text{HMAC-SHA256}_{K_i}(\mathrm{encode}(\tau_{i-1}, R_i))$, HKDF-derived K_i.